Gregory Spangenberg

Computer Vision Researcher and PhD Candidate

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Experience

DATA SCIENTIST | Eleusis Therapeutics
Oct 2021 – Jan 2022 | Remote | Contract Part-time

- Built and implemented a data ingestion pipeline to support clinical trial research

RESEARCH ENGINEER | Roth McFarlane Hand and Upper Limb Centre
Sept. 2020 – Present | St. Joseph's Health Care London

- Responsible for conducting corporate sponsored orthopedic research
- Led a study to investigate the performance of Wright Medical Group's shoulder implant
- Designed and led a study to validate a surgical navigation system for Intellijoint surgical
- Gathered and analyzed data for LimaCorporate's FDA approval of an orthopedic implant

TEACHING ASSISTANT | Department of Mechanical Engineering
Sept. 2020 – Present | Western University

- Taught "Computational Methods for Mechanical Engineers" (MME 2221B) , an introductory course on programming, data organization, processing techniques, numerical methods, and model formulation
- Led tutorial help sessions for students encountering issues with programming

STUDENT RESEARCH ANALYST | Roth McFarlane Hand and Upper Limb Centre
May – Aug. 2019 | St. Joseph's Health Care London

- Built a finite element simulation to analyze varus-valgus malpositioning of humeral total shoulder arthroplasty implants.
- Published paper: Humeral short stem varus–valgus alignment affects bone stress
- Received the Dean's Award

Research

PEER-REVIEWED ML PAPERS
- Automatic Determination of the Resection Plane for Shoulder Arthroplasty in Arthritic Humeri : A Deep Learning Model
- Automatic Bicipital Groove Identification in Arthritic Humeri for Preoperative Planning: A Random Forest Classifier Approach

PHD THESIS
- Preoperative Planning and Complication Prediction for Shoulder Arthroplasty Using Machine Learning

MITACS ACCELERATE FELLOWSHIP
- Draft Paper: Prediction of Acromial Stress Fractures from only Preoperative Imaging
- Draft Paper: Automatic Cortical Thickness Measurement of the Glenohumeral Joint using Deep Learning Segmentation and Landmarking on a Multi-Center External Dataset
- Sponsored by Industry Partner: Exactech Inc.

Skills

PROGRAMMING
Proficient:
Python • LaTeX

Experienced:
R • MATLAB • SQL

Familiar:
C++ • Fortran

LIBRARIES/FRAMEWORKS
PyTorch • Monai • Pandas • Numpy • Plotly • Dash

TOOLS/PLATFORMS
Git • Azure Cloud • Linux

LANGUAGES
English • German

Education

WESTERN UNIVERSITY
PhD Mechanical Engineering
2020 - 2025 | London, Canada
- **Collaborative Specialization in Machine Learning in Health and Biomedical Science**

WESTERN UNIVERSITY
BESc. Mechanical and Materials Engineering
2016 - 2020 | London, Canada
- **2020 Dean's List**
- **2020 Academic All-Canadian Award**

Sport

60M SPRINT HURDLER
Western University Varsity Track and Field Team
Sept 2017 - May 2023